AOGS 2024, AS65-A021, Regional Climate Downscaling and Cordex: Challenges and Prospects

Analysis of the Difference in WRF's Simulation of the East Asian Summer Precipitation in 2022 Based on PBL Schemes and CPS Combination

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1. Introduction

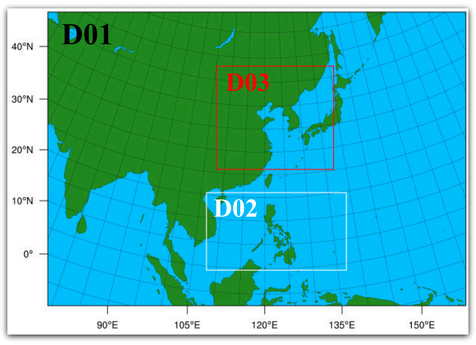
1. In **East Asia**, about half of the annual rainfall occurs in the summer, making the prediction of the **East Asian Summer Monsoon (EASM)** particularly challenging due to various influencing factors.
2. Understanding East Asian summer precipitation and improving simulation accuracy through optimal experimental design is essential. Summer rainfall in East Asia is primarily associated with **convective precipitation**.
3. The **complex topography** of this region influences the exchange of heat, moisture, and momentum between land, ocean, and atmosphere, as well as boundary mixing processes.
4. While the overall summer precipitation in East Asia in 2022 was average, the Korean Peninsula set a record for maximum hourly rainfall, and southeastern China saw its second-highest June to September precipitation since 1961.
5. This study used the **Weather Research and Forecasting (WRF)** model to analyze 2022 summer precipitation in East Asia.
6. For Cumulus Parameterization Schemes (CPS), we utilized the **KIM Simplified Arakawa-Schubert scheme (KSAS)** and the **Multi-scale Kain-Fritsch (MSKF)** schemes.
7. By employing PBL methods such as the **Yonsei University (YSU)** and **Asymmetric Convective Model (ACM2)**, the study assessed their performance and impact on weather prediction by analyzing differences in precipitation amounts.

2. Data & Methods

● Model configuration

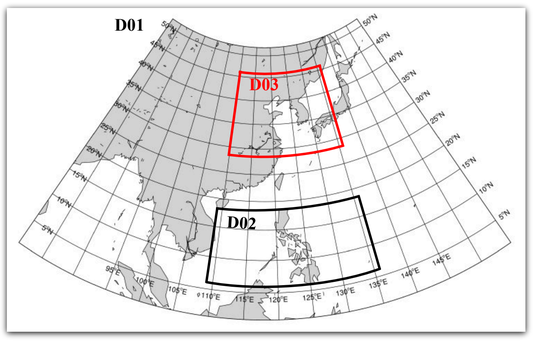
	Advanced Research WRF (ARW) v4.4			
Name	**01 YSU_KSAS**	**02 YSU_MSKF**	**03 ACM2_KSAS**	**04 ACM2_MSKF**
Horizontal Resolution (Domain)	12 km (823×570) / 4 km (826×451) / 4 km (691×601)			
Vertical Layers	35 hybrid levels			
Microphysics	WSM6			
Cumulus Parameterization	KSAS	MSKF	KSAS	MSKF
Radiation	RRTMG			
Planetary Boundary Layer	YSU	YSU	ACM2	ACM2
Land Surface	Noah LSM			
Initial & Boundary data	ERA5 (0.25° and 6 h)			
Analysis Period	0000 UTC on 1st June to 0000 UTC on 1st September 2022			

● Model Domain



1) East Asia (CORDEX phase II Domain)
   - 1 – 52 °N / 81 – 155 °E
2) Low Latitude (Philippine Sea)
   - 6 – 18 °N / 109 – 136 °E
3) Mid Latitude (Korean Peninsula)
   - 28 – 46 °N / 110 – 136 °E

● Analysis Domain

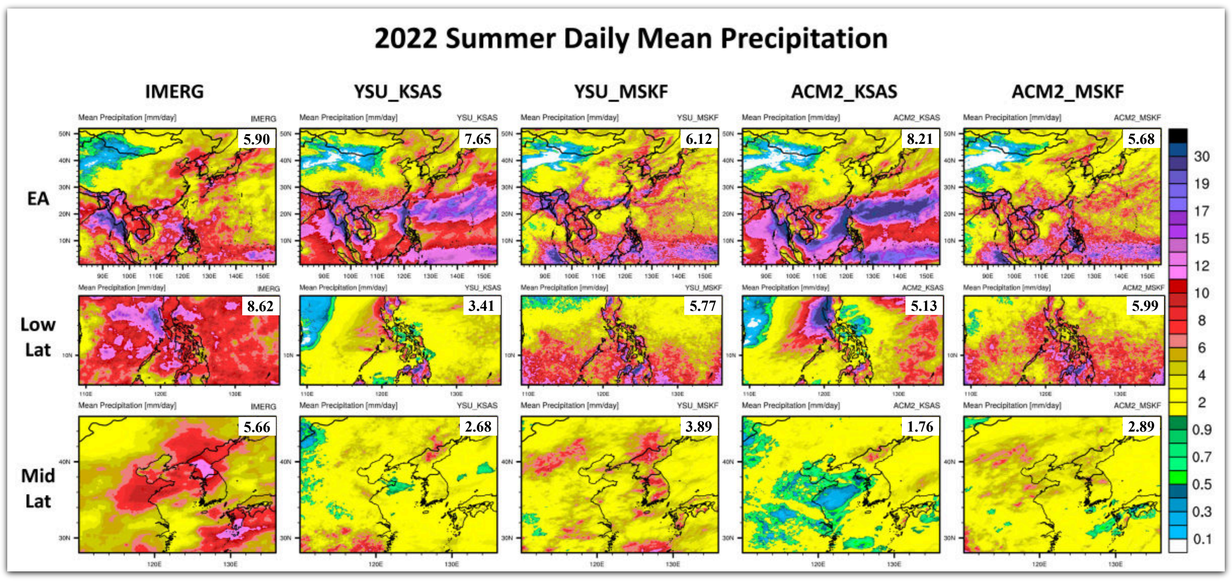


● Data

- **IMERG (Integrated Multi-satellitE Retrievals for GPM)** data is used for precipitation validation (30 min, 0.1°)
- **WRF model output** has been bilinearly interpolated to 0.1° to compare with IMERG data (both 12 km and 4 km experiment)
- **ERA5 (ECMWF Reanalysis v5)** data is used for initial and boundary condition (6 hourly, 0.25°)

3. Results

● East Asia 2022 Summer Daily Mean Precipitation



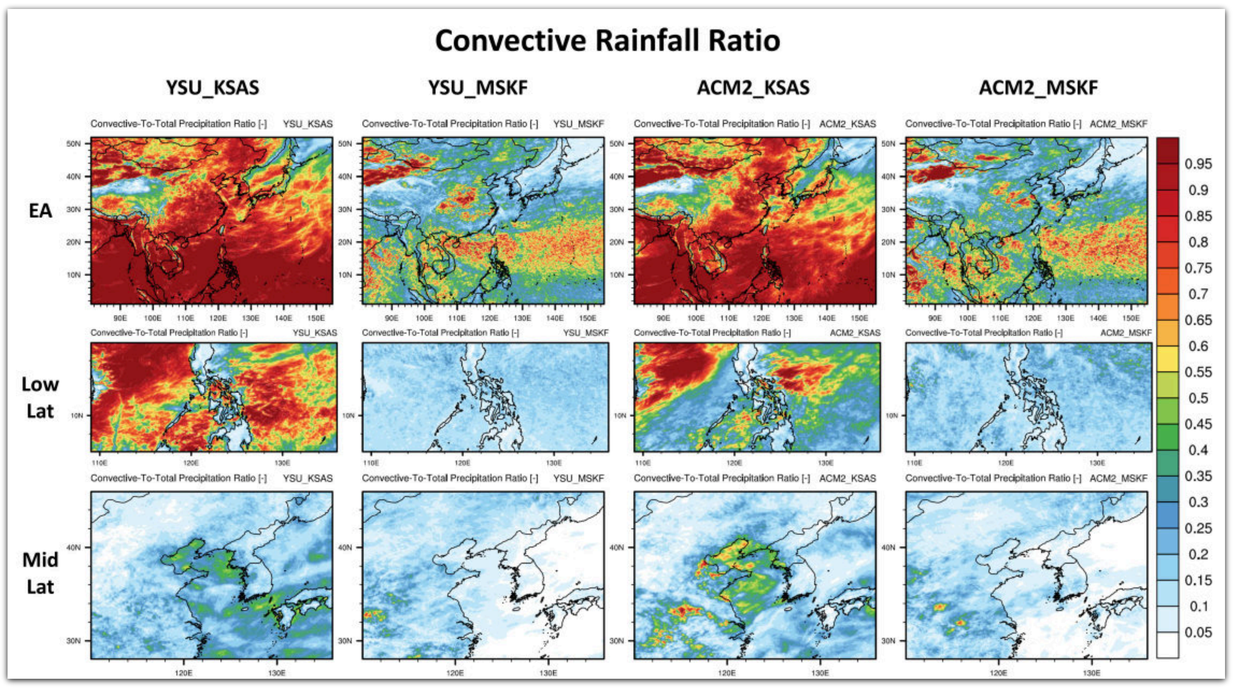
❖ Environment

- **East Asia Summer Monsoon** is well appeared in 2022 Summer
- **High precipitation** especially over the **Southeast Asia region** and **ITCZ over East Pacific**
- **Significant** precipitation over the **Philippines**
- **Moderate** precipitation in regions including parts of **Korea and Japan**

❖ Model Characteristics

- **East Asia (Domain 01)**
  - All model show different spatial variation. KSAS highly overestimated, while MSKF close to IMERG
  - **ACM2_MSKF and YSU_MSKF shows minimum bias** compared to IMERG
  - **MSKF exhibit more localized** high precipitation areas compared to KSAS
- **Low Latitude (Domain 02)**
  - KSAS shows higher precipitation intensities compared to IMERG, indicating potential underestimation in these models
  - MSKF display a more moderated and dispersed precipitation pattern, closer to IMERG but still with some variations
- **Mid Latitude (Domain 03)**
  - All model has significant variations with underestimation
  - MSKF shows a wider spread and higher intensity then KSAS
- **Overall**
  - Tend to show **higher precipitation biases**
  - **KSAS generally show higher precipitation biases** then MSKF.

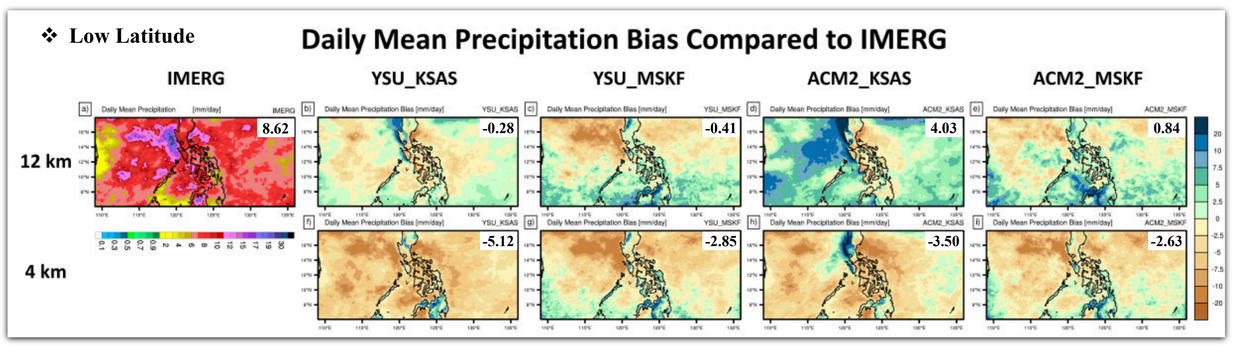
● Cumulus Parameterization Scheme Sensitivity



❖ Model Characteristics

- Both **KSAS show consistently higher CRR values**, indicating a strong bias towards convective rainfall in their simulations
- Both **MSKF** present lower CRR values, suggesting a **more balanced representation between convective and non-convective rainfall**
- **Low Latitude**
  - KSAS is well indicating the higher CRR values, as it is known (while MSKF is not)
- **Mid Latitude**
  - MSKF does not affected by PBL, but KSAS showing higher CRR with ACM2 than YSU

● Resolution Effect



❖ Model Characteristics

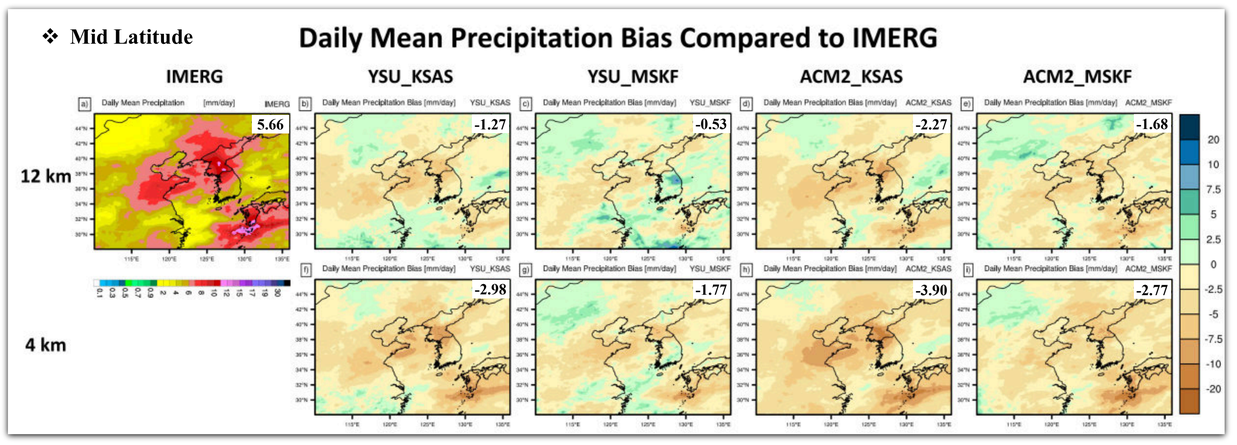
12 km

- YSU_KSAS: **lowest bias**
- YSU_MSKF: like YSU_KSAS, but with slightly increased bias
- ACM2_KSAS: **highest bias**, with significant overestimating
- ACM2_MSKF: balanced than YSU_MSKF, but larger bias

4 km

- Bias is more underestimated, **larger negative bias than 12 km**
- Bias is more underestimated, **larger negative bias than 12 km**
- Bias is significantly reduced
- Bias is more underestimated, **larger negative bias than 12 km**

- Overall, **MSKF** shows **larger negative bias on 4 km than 12 km**
- **KSAS** shows better estimation with **ACM2 on 4 km**, **YSU on 12 km**



❖ Model Characteristics

12 km

- YSU_KSAS: moderate bias
- YSU_MSKF: **best estimation** among the 8 cases
- ACM2_KSAS: larger negative bias, underestimating
- ACM2_MSKF: reduced bias compared to ACM2_KSAS

4 km

- **Bias is increased** compared to 12 km
- Continues to show **low bias**, with improved performance on some regions
- **Significant large** negative bias
- **Larger negative bias with 4 km** than 12 km

- Higher resolution does not guarantee better precipitation predicting performance (Bias: 4 km > 12 km, KSAS > MSKF, ACM2 >YSU)
- **Every model underestimates** precipitation compared to IMERG
- **KSAS is more affected by resolution than MSKF**
- **4 km experiment is more affected by PBL** than 12 km experiment

❖ Latitude & Resolution Sensitivity

- **KSAS performs better in low latitude** regions compared to mid latitude regions **at a 4 km resolution**.
- **MSKF** is not significantly affected by resolution in mid latitude regions but shows worse performance at a 4 km resolution in low latitude regions.
- There is **no consistent evidence that specific schemes consistently perform well predicting performance** regardless of variations in resolution, latitude, and PBL combinations.

4. Summary

1. **CPS with resolution**: CPS schemes are more influenced by resolution and latitude, than combination of PBL.
2. **KSAS performance**: KSAS performs **better at 4 km resolution in low latitude regions** compared to mid latitude regions.
3. **MSKF performance**: **less affected by resolution in mid latitude regions** but **worsens** at **4 km** in **low latitude** regions.
4. **Accuracy of Simulation Schemes**: The choice of CPS and PBL schemes **significantly impacts simulation accuracy**, with differences in convective-to-total precipitation ratios and biases.
5. **Inconsistent Scheme Performance**: No specific schemes consistently perform well across all resolutions, latitudes, and PBL combinations, necessitating careful selection and calibration.
6. **Importance of Scheme Selection**: **Selecting appropriate parameterization schemes and resolutions is crucial** for accurate precipitation simulations, with further research needed to improve forecasting models and predict extreme weather events.
7. **Further Study**: Analysis with synoptic patterns related each scheme's properties.

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